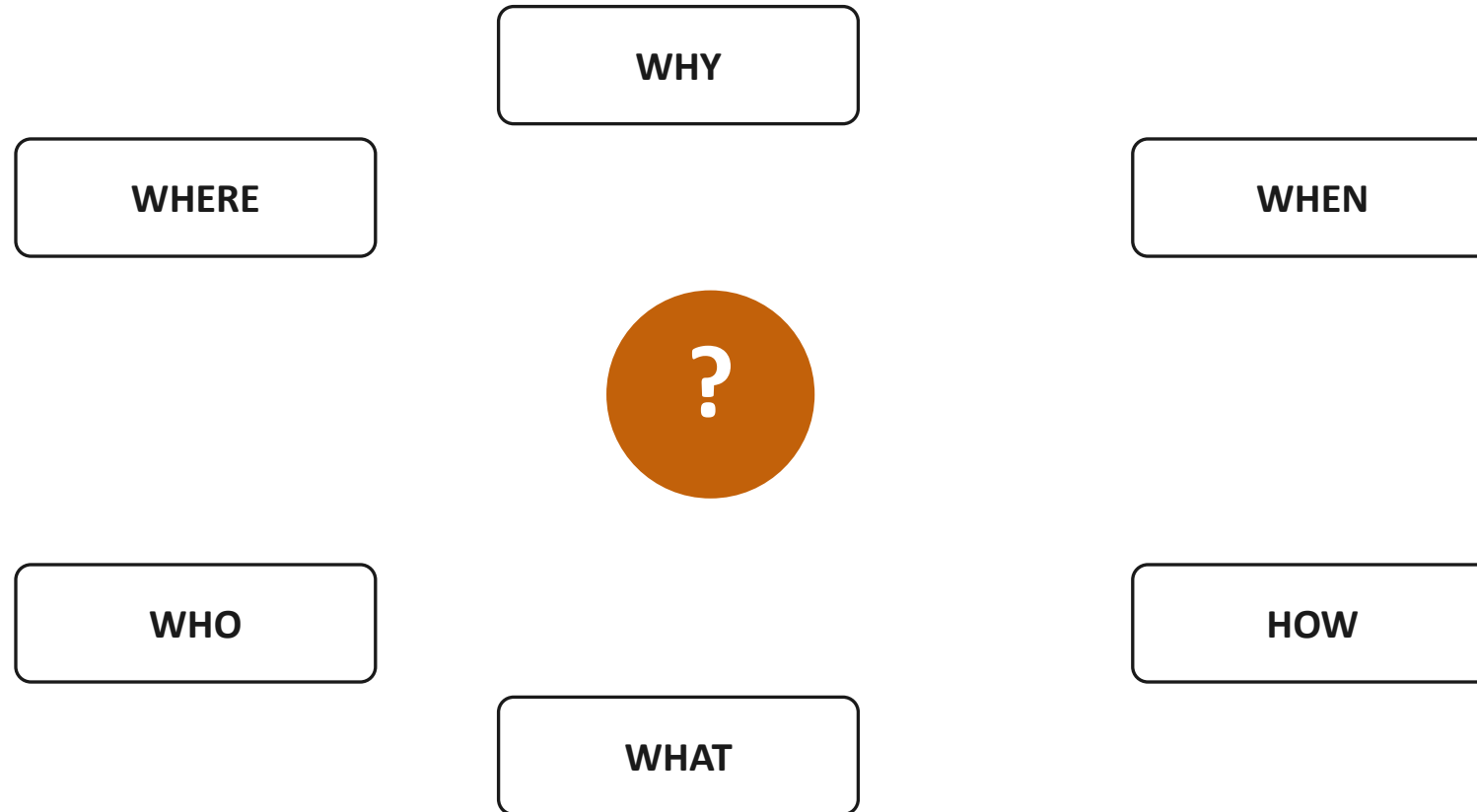


IDRM.I

Integrated Disaster Response Management · for India

Minimum Viable Product

IDRM — the questions we answer



IDRM — why: disaster types

India is prone to a wide range of disasters — social, economic and environmental in impact.

- **Floods** — most common, annual monsoon flooding.
- **Droughts** — irregular rainfall; agriculture & livelihoods.
- **Cyclones** — coastal storms, surges, infrastructure damage.
- **Earthquakes** — Himalayan & seismic zones.
- **Landslides** — hilly regions during heavy rain.
- **Heatwaves** — deadly summer temperatures.
- **Wildfires** — dry/forested regions; biodiversity loss.
- **Tsunamis** — undersea quakes; coastal inundation.
- **Urban flooding** — drainage overwhelmed in cities.
- **Cold waves** — northern winters; health & agriculture.
- **Avalanches** — high-altitude Himalayas in winter.

IDRM — vision & mission

- **Vision** — a disaster-resilient world where communities, institutions and responders act early, respond faster and recover stronger — saving lives and livelihoods.
- **Mission** — to integrate people, processes, data and technology on a unified platform enabling collaboration, situational awareness and informed action across the lifecycle.
- **Principles** People-Centric · Integrity & Trust · Interoperability · Timeliness · Innovation · Sustainability.

IDRM — stakeholders

- **Citizens & communities** — report, request help, receive alerts, give feedback.
- **Volunteers & community groups** (NGOs / CBOs) — force multipliers in relief.
- **Emergency services** — Police, Fire, Medical, SDRF, NDRF, Coast Guard, Home Guards.
- **District administration & EOCs** — command, coordination, allocation, oversight.
- **State & national agencies** — NDMA, MHA, NDRF, IMD, ISRO, MoHFW, BRO, Railways.
- **Partners & external systems** — weather/IMD, satellite/ISRO, health, UN, media, crowd sources.

IDRM — personas

- **Citizen** — reports needs, tracks status, receives alerts. Needs simplicity & language support.
- **Volunteer** — registers availability & skills, responds to tasks, shares updates.
- **Emergency responder** — acts on alerts, executes response, shares real-time status.
- **Coordinator / District admin** — triages, allocates resources, issues directives, communicates.
- **Administrator** — manages users, roles, configuration and audit oversight.
- **Government official** — dashboards & reports for decisions and accountability.

IDRM — business domains

- **Incident Management** — report, classify, track, escalate.
- **Resource Management** — inventory, fleet, shelters.
- **Volunteer Management** — register, skills, assignments.
- **GIS & Mapping** — maps, layers, routing.
- **Communication** — alerts across channels.
- **Analytics** — reports, KPIs, dashboards.

IDRM — the disaster-management cycle

Prevention & Preparedness

- Risk & vulnerability assessment
- Planning, SOPs, regulations
- Training & field exercises
- Early-warning readiness
- Strengthening resources

Response & Intervention

- Decision-making & mobilization
- Coordination of authorities
- Search & rescue, first aid
- Damage assessment
- Communication to public

Restoration & Recovery

- Relief & temporary housing
- Financial & social support
- Restore public services
- Psychological support
- Evaluate & improve plans

IDRM — response phases

Immediate response

- I · Search & rescue operations
- II · First-aid & medical care
- III · Mitigate induced-phenomena impact
- IV · Provide emergency supplies
- V · Provide emergency shelters
- VI · Post-event building inspections
- VII · Immediate financial relief

Mid-term response

- I · Semi-permanent housing
- II · Suspension of tax liabilities
- III · Housing assistance for restoration
- IV · Suspension of insurance contributions
- V · Extensions for tax declarations

IDRM — guiding principles

- Comprehensive & proactive planning
- Preparedness → Response → Recovery
- Post-event evaluation & learning
- Resilience & sustainability
- Flexible & adaptive strategies
- Data security & privacy
- Transparency & information sharing
- Inclusive & equitable response
- Technology integration & resource optimization
- Coordination & collaboration across stakeholders

Analogy: the anatomy of the human body

IDRM behaves like a living system — it senses & signals, and it moves resources.

The neural network — communicates

- **Information** — the brain orchestrates organs via electric **signals**
- → the **Information / Signal Highway**

The circulatory network — transports

- **nutrients / oxygen** → healthcare, food & goods
- **hormones** → reserved forces
- **toxins away** → risk mitigation
- **anti-bodies** → preservation, protection & feedback
- → the **Logistics / Transportation Highway**

IDRM — the two highways

Information / Signal Highway

situational awareness · alerts · knowledge · feedback

Logistics / Transportation Highway

resources · relief supplies · responders · money pooling

IDRM.I web-portal

Disaster Response Management for India — a web-application portal that serves as:

- **Dashboard interface** — for all interested parties (requestors, volunteers, providers, governance, audit) via map annotations and visual charts.
- **Knowledge Management System** (CMS/KMS) — information via wikis, chatbot, SMS, preferably with context tracking.
- **Money-pooling portal** — for needy individuals, causes and service activities, with accountability.
- **Analytics platform** — logging & tracking all services, with anonymized identity-aware services and privacy.

IDRM.I — the five C's

- Capacity Planning
 - Capability Building
 - Communication
 - Collaboration
 - Community outreach
- Privacy-controls
 - Accessibility
 - Internationalization
 - (locale support)

IDRM — services & processes

Services

- Services enumeration & tracking
- Dynamic collaboration
- Transparent communication

Processes

- Feedback
- Escalations
- Memos
- Audit
- Mis-use & anomaly detection
- Privacy preservation & protection

IDRM — how we deliver (SDLC)

- **Idea** → discovery, feasibility, value → concept note
- **PRD & Architecture** → requirements, design → specs
- **Development** → build to standards, versioned → working software
- **Testing** → functional, performance, UAT → quality sign-off
- **Deployment** → CI/CD, go-live, rollback → production
- **Operations** → monitor, backup, support, SLA → stable system
- **Continuous Improvement** → feedback, tuning, roadmap → better outcomes

Part II — the technical deep-dive

architecture · data · security · quality · operations

IDRM — architecture (a modular monolith)

One Python 3.12 application. Simple to run, tidy inside, ready to grow.

- **Web UI** — Flask + Jinja2 + Bootstrap 5 (server-rendered).
- **APIs** — FastAPI + Pydantic (validated, OpenAPI 3.1).
- **Background work** — threading / ThreadPoolExecutor + APScheduler (in-process).
- **Data** — PostgreSQL + PostGIS (primary) · Redis (cache) · object storage.
- Evolution Prototype → **MVP** → Production Monolith → Enterprise FFP.

IDRM — the eight layers

- 1 · User & Interaction
- 2 · Core Functional
- 3 · Business Services
- 4 · Platform Services
- 5 · Data & Integration
- 6 · Intelligence & Decision Support
- 7 · Infrastructure & Operations
- 8 · Security, Governance & Compliance

IDRM — modules & components

- Authentication & Authorization
- Incident Management
- Resource Management
- Volunteer / Responder
- GIS & Mapping
- Notification / Alerts
- Reports & Analytics
- Document / Media
- Audit & Activity
- + Administration & Integration services

IDRM — how a request flows

- **Request** → routing → middleware (CORS, auth, log) → **authentication** (JWT)
- → **authorization** (RBAC) → validation (Pydantic) → controller → business module
- → **data access** (SQLAlchemy) → PostgreSQL / PostGIS / Redis → response
- **Background:** task → queue → worker thread → execute → notify → complete (APScheduler for schedules).

IDRM — technology stack

- **Python 3.12** — one language.
- **Flask · Jinja2 · Bootstrap 5** — web UI.
- **FastAPI · Pydantic** — APIs.
- **threading · APScheduler** — jobs.
- **Miniconda** — env & GIS/GDAL.
- **SQLAlchemy · Alembic** — ORM & migrations.
- **PostgreSQL · PostGIS** — data + maps.
- **Redis** — cache/sessions (optional).
- **Gunicorn · Uvicorn · NGINX** — serving.
- **pytest · Ruff · Black · mypy** — quality.

IDRM — repository structure

One repository; each module slices across models, schemas, services, apis, routes.

- **app/** models · schemas · services · apis · routes · templates · static · tasks · utils
- **migrations/** Alembic
- **tests/** unit · integration
- scripts/ · run.py · requirements.txt · environment.yml · .env.example

IDRM — database architecture

- **PostgreSQL** — the system of record.
- **PostGIS** — spatial types & indexing (GiST); proximity queries.
- **Redis** — cache, sessions, rate-limit counters (optional).
- **Object storage** — MinIO (S3-compatible) or local/S3: documents, media, backups.

IDRM — entity model (18 tables)

- **Access & Administration** users · roles · organizations · locations
- **Incident Management** incidents · incident_types · incident_updates · alerts
- **Response Management** responders · resources · resource_types · deployments · tasks
- **Data & Content** documents · media · geospatial_data
- **Communication & Audit** communication_logs · audit_logs
- UUID keys · created_at/updated_at · PostGIS geometry · immutable audit

IDRM — GIS & map services

Python + GDAL + PostGIS — no separate map server for the MVP (GeoServer optional, future).

- **Map Annotator** — service request ID, geo-location.
- **Map Summarizer** — counts; region-wise & category-wise.
- **Map Work-flows** — collection, clustering, routing & optimization.
- EPSG:4326 · GiST indexes · ST_DWithin.
- Layers: locations, boundaries, hazard zones, assets, routes, areas.
- Leaflet maps · caching for re-rendering.

IDRM — the service request

The core record — maps to the incidents entity.

- Service Request ID
- Service Type
- Requestor ID
- Provider ID
- Geo location (lat, lon)
- Description / comments
- Status
- Tasks & timeline tracking

IDRM.I requirements

- **IAM** — AAA (OAuth2, SAML, JWT) with **RBAC** for all user types.
- **Privacy-controls** — Privacy Preservation & Protection (PPP).
- **Map, location- & identity-aware services** — dashboards, routing, export, caching.
- **Data-warehousing** — analytics & activity logging.
- **Chatbot** — capture requests & feedback; social-media integration.
- **Money pooling** — enabling service, social & charitable activities.

IDRM — security

- **Layers** — application · data · infrastructure · operational.
 - **Principles** — Zero Trust · Defense in Depth · Least Privilege.
 - **Workflow** — authN (OIDC/MFA) → JWT → RBAC → audit.
- HTTPS/TLS · input validation · CSRF/XSS/SQLi protection.
 - Rate limiting · security headers · immutable audit.
 - ISO 27001 · NIST CSF · OWASP · DPDP Act 2023

IDRM — threat model

OWASP + STRIDE · assets · trust boundaries · risk = likelihood × impact.

- **A01** Broken access → RBAC, least privilege
- **A02** Crypto failures → TLS, encryption, secrets
- **A03** Injection → validation, ORM
- **A05** Misconfig → hardening, patching
- **A06** Vulnerable components → scanning
- **A07** Auth failures → OIDC, MFA, logout
- **A09** Logging failures → central logs, alerts
- **A10** SSRF → egress filtering, URL validation

IDRM — testing

- **Unit ~60%** · API ~20% · Integration ~15% · E2E ~5%
- + database · performance · security · contract · UAT.
- **Coverage $\geq 80\%$** ($\geq 70\%$ gate). Tools: pytest, Ruff, Playwright, k6, OWASP ZAP.
- Shift left — test early, run in CI/CD on every change.

IDRM — CI/CD

- **Commit** → Lint & Format → Type Check → Test → Coverage → Package → **Deploy**
- **GitHub Actions**; fail-fast. Release gates: lint, tests, coverage, no high/critical vulns, smoke checks.
- DORA: deployment frequency · lead time · change-failure rate · MTTR.

IDRM — deployment (Ubuntu)

- **Standalone Ubuntu 22.04 LTS** — ~4 vCPU / 16 GB / 500 GB SSD, static IP.
- NGINX (TLS) → Gunicorn (Flask) + Uvicorn (FastAPI) + in-process jobs; PostgreSQL/PostGIS, Redis; systemd.
- **Ports:** 22 (SSH) · 80 → 443 · rest blocked (UFW).
- **Hardening:** SSH-key, Fail2Ban, Let's Encrypt, secrets in .env, patching.

IDRM — observability & SLO

- **Logs** — Loki/ELK.
- **Metrics** — Prometheus/Grafana.
- **Traces** — OpenTelemetry/Tempo.
- **Alerts** — Alertmanager.

- **SLO/SLA**
 - Availability $\geq 99.5\%$
 - API p95 < 800 ms · error < 1%
 - Incident ack < 5 min · P1 < 1 hr

IDRM — disaster recovery

- **Backup** — DB (daily + WAL, 30 d) · files · configs · system — local + off-site.
- **RPO / RTO** — web & DB 1 hr / 2 hr · files 4 hr / 4 hr.
- **Runbook** — Detect → Assess → Failover → Recover → Validate → Communicate → Review.
- 3-2-1 rule · quarterly restore & semi-annual failover drills · pg_dump, borgbackup, rsync.

IDRM — roles & ownership

- **Roles by SDLC**

- Product/BA · Architect
- Backend/Frontend · Data/GIS
- Security · QA · DevOps

- **Ownership (RACI)**

- APIs → API Lead · Database → DBA
- Security → Security Eng.
- Testing → QA · CI/CD → DevOps

IDRM — the 10-year roadmap

- **1 • Foundation** (Yr 0–1) — core, RBAC, maps, stable on-prem.
- **2 • Stabilization** (Yr 1–2) — performance, HA, governance, DR.
- **3 • Integration** (Yr 2–3) — APIs, remote sensing, IoT feeds.
- **4 • Intelligence** (Yr 3–5) — AI/ML prediction, decision support.
- **5 • Autonomy** (Yr 5–7) — automation, auto-dispatch, chatbot.
- **6 • Resilience 2.0** (Yr 7–10) — self-healing, cross-border interop.

IDRM — the big picture

- **People** — citizens, volunteers, responders, officials, agencies, partners.
- **Capabilities** — incident · resource · volunteer · GIS · communication · analytics.
- **Foundation** — one Python app · one DB with maps · secure & observable · ready to grow.

One platform. Every stakeholder. Resilient communities.

IDRM.I references

- [Sahana](#) — disaster-management software
- [GeoCove](#) · Disaster Risk Reduction Toolkit
- [Rasa](#) — NLP/NLU chatbot
- [Meta](#) — Data for Good
- **Standards** — WCAG 2.2 AA · ISO 27001 · ISO 22301
- DPDP Act 2023 · OpenAPI 3.1 · OWASP
- ISO/IEC/IEEE 29148 · 42010 · NIST CSF

Prepared today. Protected tomorrow.

Technology in service of humanity.

IDRM.I